Fiscal Year:	FY 2011	**Task Last Updated:**	FY 09/24/2011
PI Name:	Beckingham, Kathleen M. Ph.D.		
Project Title:	The role of mechanoperception in gravity responses of the model organism Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	77005	**Congressional District:**	25
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Task Description:	The fruit fly Drosophila melanogaster offers unique opportunities for genetic and molecular studies of cellular processes common to all multicellular organisms. In order to learn more of the pathways activated by the force of gravity, we have isolated a series of Drosophila mutants that have altered behavioral responses to gravity. These and other Drosophila mutants will be used to explore the signaling pathways underlying one the most marked changes associated with exposure to microgravity - loss of muscle tissue. Specifically the studies will address the issue of whether mechanosensory neurons and associated central brain circuitry regulate muscles responses to altered gravitational force. A powerful technique has been developed within Drosophila that allows conditional inactivation of targeted neurons. This technique will be used to inactivate the neurons under study and to determine whether this loss of neural input alters the responses of the muscles to hypergravity. Quantitative PCR will be used to examine muscle responses. A selection of genes indicated by other studies to show altered expression in response to altered gravitational force will be examined.		

	These studies should provide unique information on the regulation of muscle metabolism in response to changes in gravity. NRA NNH08ZTT003N seeks to identify studies aimed at understanding signaling pathways associated with mechanoperception. The studies proposed are focused directly upon this goal. NASA has programmatic interest in understanding the adaptation of living things to change in gravitational force. The studies presented will provide significant insight into these adaptive processes. The major team member associated with these studies is senior research technician Rebecca A. Simonette.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The studies under this project address the question of how muscles respond to increase in mechanical force by growth or decrease in mechanical force by atrophy. In addition to providing insight into the effects of hypo-and hyper-gravity on muscle mass these studies are highly applicable to an understanding of medical situations that occur on Earth. In particular, the muscle atrophy experience from confinement to bedrest almost certainly involves similar regulatory mechanisms. By uncovering the mechanisms by which muscles sense and respond to altered mechanical this research will have significant impact upon human health.
Task Progress:	Muscles are highly responsive to mechanical load. Muscles that are heavily used undergo growth whereas muscles that are not subjected to mechanical work atrophy and become weak. Degeneration of the muscles is thus a common problem for astronauts under the low gravitational force experienced in space. A major question concerning these observations is – how do the muscles sense their mechanical load and how do they respond to it? One hypothesis is that neurons within the muscles specialized to sense the force exerted on the muscles somehow provide feedback to the muscles initiating either growth or atrophy. The research performed under this project was aimed at investigating this possible relationship between mechanosensory neurons and muscle metabolism. The experiments were performed in the model organism Drosophila melanogaster, a small fruit fly. This organism has been studied intensively for about 100 years and as a result, many research tools are available for studies of muscle function in this organism that have not been generated for other animal species. The project grew out of our previous work to identify genes that contribute to responses to gravitational force. During the course of this funding, we have pursued molecular and behavioral studies of two of these genes, which we have named yuri gagarin (yuri) and jim lovell (lov) in honor of our NASA funding and two heroic space pioneers. We have established that the yuri gene encodes a protein that interacts with a key component of the muscle contraction apparatus. This protein, termed Tropomyosin, has a major role in stabilizing one of the rod-like structures in muscles that move during the contraction process. Using a specialized technique termed TAP-tagging, we have shown that the protein encoded by Yuri binds directly to the Tropomyosin protein (see publication Texada et al., 2011, in below Bibliography). Given that we know that mutations to the yuri gene affect locomotor responses to gravity (1), these findings suggest that yuri expression may be regulated by mechanosensory neurons. Our surprising finding that yuri is required for attachment of the centriole (basal body) to the spermatid nucleus (see publication Kracklauer et al., 2010, in below Bibliography), also suggests an alternative route whereby yuri may influence gravitational responses. All mechanosensory neurons in Drosophila have cilia that grow out of a modified centriole and are essential for the mechanosensory signal transduction. Thus it is possible that yuri is a component of the centriole/cilia complex in these neurons and may act in gravitational sensing at the level of the neuron. The second gene studied during this funding period (lov) has characteristics that suggest it also acts in mechanosensory neurons, specifically a particular class termed chordotonal neurons. lov encodes a neural specific transcription factor of the BTB-POZ domain family and we have established that Lov protein is expressed in a subset of chordotonal neurons associated with muscles involved in locomotion. Given this association of the Lov-expressing neurons with locomotion, we set out to determine whether the Lov neurons actually control muscle activity. To investigate this possibility, we used one of the tools especially developed in Drosophila melanogaster. This is a genetic method that allowed us to express Lov in cell types that do not normally express the protein. The actual experiments performed involved expressing Lov in chordotonal neurons that do not normally express Lov, the rationale being that if Lov controls muscle activity and thus helps produce coordinated muscle movement by being expressed in particular chordotonal neurons, locomotion should be disrupted in some way if Lov is expressed in different chordotonal neurons. Initially we chose to apply this approach during the larval stage of development since the nervous system is much better understood during this period. Strikingly, when Lov is misexpressed in this way, the coordinated muscle movements of the larvae are dramatically affected. The larvae develop from embryos in eggshells prepared by their mothers. At the end of embryonic development the newly formed larva must perform a particular series of muscle movements in order to hatch from the eggshell. The larvae with misdirected expression of Lov are incapable of performing these muscle movements. The hatched larvae move by a “caterpillar like” locomotion that involves highly controlled waves of contraction along the body from the back to the front. When the Lov misexpressing larvae are physically removed from their eggshells and set on a smooth surface they fail to generate these coordinated waves of contraction and cannot move from any position in which they are placed. Ultimately the larvae starve since they are also incapable of the movements needed for feeding. We have also misexpressed Lov during the adult stage of Drosophila development. The adults form in pupal cases generated by the larvae at the end of larval development. When Lov is misexpressed so as to affect adult locomotor neurons, most adults are incapable of eclosing from their pupal cases, again because of failure to make the necessary coordinated movements. If helped from their pupal cases, the adults are highly uncoordinated, falling frequently, failing to fly and dying rapidly due to an inability to forage effectively for food. All of these findings suggest the neurons expressing Lov protein have major roles in coordinating locomotion. Other studies have indicated that much of this coordination is executed by the neurotransmitter octopamine (the invertebrate equivalent of nor-adrenalin) (2). An exciting recent finding is the discovery that lov regulates the production of octopamine, presumably by effecting the terminal differentiation of octopaminergic neurons. Future studies are aimed at investigating the connection between lov and bioamine producing neurons in more detail. REFERENCES 1. Armstrong, J. D., Texada, M. J., Munjaal, R.V. Baker, D. A. and Beckingham, K. "Gravitaxis in Drosophila: a

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Bibliography Type:	Description: (Last Updated: 03/14/2017)
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